Lam Fung Academy

SSPA Mock Exam (4) · Primary 5 Mathematics

Scope: Year-round Comprehensive (Lessons 1–28) — Multiple digits • Estimation • Area • Fractions • Decimals • Algebra • Equations • Circles • Solid sections • Origami patterns

Exam time: 75 minutes Full score: 100 points

ame: ______________ date: ______________ Simulation 1 score: ______ Simulation 2 score: ______ Fraction: ______ / 1(

————— Part A: Multiple choice questions —————

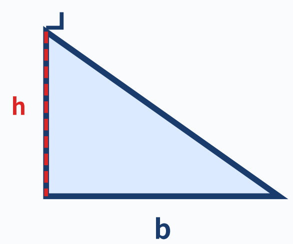
Part A - Multiple Choice Questions

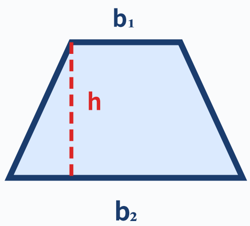
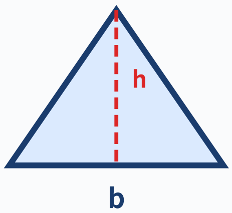
15 questions × 2 points = 30 points

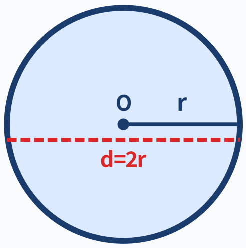
Instructions: Choose the correct answer and fill in the letter of the answer in the space to the right of each question. There is only one correct answer for each question. Pay attention to the traps hidden in the questions!

Exam tips (SSPA must read)

① Do the questions you know first, don’ t get stuck on one!
② Check the unit for each question (cm vs cm^2 vs cm^3)!
③ Geometry questions: If there is an elevation in the picture, use the height. If there is no elevation, use the formula to find it!
④ Application question: Write an answer sentence! Points will be deducted for not writing steps!
⑤ 5 minutes left: Check if the MC has been filled in and if the unit is correct!

question number	Question	fraction
1	How many "zeros" do you need to pronounce the number 70,600,050 in Chinese? A. 0 pieces B. 1 piece C. 2 pieces D. 3 pieces	___ / 2 points
2	In the number 8,305,206, what is the digit of "3"? What is its value? A. Millions, 3,000,000 B. Hundreds, 300,000 C. Tens, 30,000 D. Thousands, 3,000	___ / 2 points
3	Round 4,872,593 to the nearest hundred thousand digits. turn out? A. 4,800,000 B. 4,870,000 C. 4,900,000 D. 5,000,000	___ / 2 points
4	▲Triangle: base 16cm, height 9cm  Triangle base length 16cm, height 9cm. Area = ? A. 144 cm^2 B. 72 cm^2 C. 48 cm^2 D. 25 cm^2	___ / 2 points

5	▲ Trapezoid: hypotenuse 11cm (trap! You don’ t need a hypotenuse to calculate the area)  Trapezoid: upper base 7 cm, lower base 13 cm, hypotenuse 11 cm, height 6 cm. When finding the area, which of the following numbers**does not require**to be used? A. 7 cm B. 13 cm C. 11 cm D. 6 cm	___ / 2 points
6	Calculate $\frac{3}{4} + \frac{1}{6}$ = ? Which of the following is the correct answer? A. $\frac{4}{10} = \frac{2}{5}$ B. $\frac{11}{12}$ C. $\frac{9}{12} + \frac{2}{12} = \frac{11}{12}$ D. $\frac{3+1}{4+6} = \frac{4}{10}$	___ / 2 points
7	Calculate 0.35 × 0.4 = ? Where should the decimal point be placed? A. 0.014 B. 0.14 C. 1.4 D. 14	___ / 2 points
8	Simplify: $4a + 7 - 2a + 3 =$? A. $2a+4$ B. $2a+10$ C. $6a+10$ D. $6a+4$	___ / 2 points
9	Solve the equation: $2x + 5 = 3x - 2$，$x =$? A. 3 B. 5 C. 7 D. −3	___ / 2 points
10	▲ Parallelogram: Base 2m, height 80cm (Trap! Units are inconsistent)  The parallelogram has a base 2m long and a height 80cm. Area = ? A. 160 cm² B. 16,000 cm² C. 1.6 m² D. 16 m²	___ / 2 points

question number	Question	fraction
11	"The $\frac{2}{3}$ of a number is equal to 18. What is this number?" Which of the following equations is correct? A. $x+\frac{2}{3}=18$ B. $\frac{2}{3}x=18$ C. $x-\frac{2}{3}=18$ D. $18\times\frac{2}{3}=x$	___/2 points
12	There are 80 pieces in a pack of sugar. I ate |||SEP|||, and then gave the remaining $\frac{1}{4}$ to my sister. How many pills does the sister get?**A. 20 capsules** $\frac{1}{3}$ B. 15 capsules C. 25 capsules D. 40 capsules C. 25 capsules D. 40 capsules	___/2 points
13	▲ Circular flowerbed: radius 7m  A circular flowerbed, radius = 7 m. What is the approximate circumference of a circle? (Take π = 3.14) A. 21.98 m B. 43.96 m C. 153.86 m D. 49 m	___/2 points
14	If you cut a cube diagonally, what is the cross-section shape? A. Square B. Rectangle C. Triangle D. Trapezoid	___/2 points
15	A rectangular origami pattern has 6 connected squares. What three-dimensional shape can be formed by folding it? A. Cube B. Cuboid C. Cylinder D. Unable to form a solid	___/2 points

————————— Part B: Short answer calculation questions —————————

Part B - Short Questions/Calculation Questions

10 questions × 4 points = 40 points

Instructions: Write the complete calculation steps and answers in the answer area below each question. Answers must be reduced (if applicable) and accompanied by units.

question number	Question	fraction
1	(a) Write in Arabic numerals: nine thousand three hundred and sixty thousand. (b) Round the above numbers to the nearest million.	___/4 points
2	Estimate: 4,876 + 3,209 ≈ ? (Round to the nearest thousand, using the "≈" symbol)	___/4 points

3	Triangle base = 18 cm, height = 7 cm. Area = ? Write down formulas and calculation procedures.	___ / 4 points
4	The upper base of the trapezoid is 8 cm, the lower base is 14 cm, and the height is 5 cm. Area = ? Write down formulas and calculation procedures.	___ / 4 points
5	Calculation: $\frac{2}{3} + \frac{3}{8}$ = ? Write down the general steps and final answer (the simplest fraction).	___ / 4 points

question number	Question	fraction
6	Calculation: $1\frac{2}{5} \times \frac{3}{4}$ = ? Write out: mixed numbers → improper fractions → reduction → multiplication → answer.	___ / 4 points
7	Calculate the following decimal operations: (a) $0.6 \times 0.35 = ?$ (b) $1.25 \div 0.5 = ?$	___ / 4 points
8	Solve the equation: $3(2x - 1) = x + 7$ Write out: remove brackets → move terms → merge → solve all steps.	___ / 4 points
9	The radius of a circle is 10 cm. Find: (a) Circumference = ? (Take $\pi = 3.14$) (b) Circle area = ? (Take $\pi = 3.14$)	___ / 4 points
10	The side length of the cube is 5 cm. Find: (a) Volume = ? (b) Surface area = ?	___ / 4 points

Part C - Application Problems

4 questions × 7.5 points = 30 points

Instructions: Each question must list all calculation steps (formulation, solution, verification) and write a complete answer. The answer must be in units or reduced to the simplest form.

Question 1 (7.5 points) – Multi-digit + estimated synthesis Cross-topic ___ / 7.5 points

Here are the population data for the three cities:
City A: 2,856,400 people
City B: 1,439,800 people
City C: 3,705,200 people

(a) Approximate the population of each of the three cities to 100,000 people. (1.5 points)
(b) Use approximations to estimate the total population of the three cities. (2 points)
(c) Calculate the actual total population of the three cities using exact values. (1.5 points)
(d) How much does the estimated total population differ from the actual total population? Is the estimate reasonable? (2.5 points)

Answer area - please write all calculation steps and complete answer sentences:

Question 2 (7.5 points) – Area of combined shapes (trap question) Cross-topic ___ / 7.5 points

The picture below is made up of a rectangle and a triangle.
Rectangle: length 12 cm, width 8 cm.
Triangle: Spliced on top of the rectangle, the base is the same as the length of the rectangle (12 cm),**Bevel length 10 cm**, 6 cm high.

(a) Find the area of the rectangle. (1 point)
(b) Find the area of the triangle. (Note: the hypotenuse is not high!) (2 points)
(c) Find the total area of the entire combined figure. (1.5 points)
(d) If the width of the rectangle is increased by 50% and the triangle remains unchanged, what is the new total area? (3 points)

Answer area - please write all calculation steps and complete answer sentences:

Question 3 (7.5 points) – Mixed Fractions + Decimals + Equations Cross-topic ___ / 7.5 points

Mom bought 4 boxes of juice, each box is 1.25 liters (L).
She poured all the juice into a large bottle.
On the first day, I drank all the juice $\frac{2}{5}$ 。
The next day, I drank another 1.5 L from the remaining juice.

(a) How many liters of total juice are there? (1 point)
(b) How many liters did you drink on the first day? How many liters are left? (2 points)
(c) How many liters of juice are left after the second day? (2 points)
(d) What fraction of the original juice is left in the end? (expressed as a simple fraction) (2.5 points)

Answer area - please write all calculation steps and complete answer sentences (with unit L):

Question 4 (7.5 points) — Graph + Equation Comprehensive Ultimate Challenge Cross-topic ___ / 7.5 points

The length of a rectangle is 4 cm more than its width.
Given perimeter = 40 cm.

(a) Let the width be w cm, expressing length algebraically. (1 point)
(b) List the equations based on the perimeter formula and solve the equations to find w. (2.5 points)
(c) Find the length and then the area of the rectangle. (2 points)
(d) If the length and breadth of this rectangle are each increased by 2 cm, by how many cm^2 will the area of the new rectangle be increased? (2 points)

Answer area - please write all steps, equation derivation and complete answer sentences:

Part C continued answer area: If there is insufficient space for your answer above, please continue here. Clearly indicate the title number.

Rough Work: The content on this page will not be rated.

—Complete the entire volume—

Total score:________/ 100 points

Part A ___/30 + Part B ___/40 + Part C ___/30 = Total score ___/100

Signature of marking teacher: __________ Date: __________

Exam Tips (Mock 4 - Ultimate Reality, 75 minutes, 100 points, comprehensive throughout the year):

- Part A MC 15 questions (15-18 minutes): Question 2 digit value counting from the right, Question 5 Hypotenuse ≠ height, Question 10 Unify the units first, Question 12 Note that the "remaining 1/3" standard quantity is "remaining" not "all", Question 13 Distinguish circumference vs circle area
- Part B Calculation 10 questions (25-28 minutes): Question 1 Chinese → Be careful with numbers 0, Questions 5-6 Answers must be simplified after passing points, Question 7 Number of decimal multiplication digits must be counted, Question 8 Remove brackets and assign negative signs
- Part C Application 4 questions (28-30 minutes): Question 2 Hypotenuse ≠ Height, Question 3 Reference Quantity Tracking (Drink 2/5 first, and leave 3/5), Question 4 Two-way Calculation of Graphs + Equations
- Simulation 4 is more comprehensive than Simulation 1/2/3: new questions on circles, three-dimensional sections, and origami patterns are added
- Make good use of 75 minutes: 18 minutes for Part A → 28 minutes for Part B → 24 minutes for Part C → 5 minutes for inspection
- Simulation 1 score ___/100 → Simulation 2 score ___/100 → Simulation 4 score ___/100 (Track the progress curve!)

Appendix: Simulation 4 Post-examination Supplementary Practice

16 questions·For after-class consolidation

Instructions: The following questions are aimed at the areas where you are most likely to lose points in the full-year simulation of Simulation 4. Please complete them after the exam.

morenumber of digitsandestimation（total 3 question）

#	Question	Difficulty
S1	Use 5, 0, 0, 9, 0, 0, 2, 8 to form the largest eight-digit number that can read "two zeros".	🌿
S2	In a nine-digit number, the billion digit is 3, the tens and millions digits are both 0, and the hundreds of thousands digit is 7. There are only two "zeros" read out, and the remaining numbers are different and as large as possible. What is this number?	⛰
S3	Approximate 23,568,917 to (a) tens of thousands (b) millions. Which one is closer to the original number?	🌿

areaspecial topic（total 4 question）

#	Question	Difficulty
S4	The parallelogram has a base of 15 cm and a height of 8 cm. Area = ?	🌱
S5	Trapezoidal area 90 cm², height 6 cm. The upper sole is 4 cm less than the lower sole. Find the upper and lower bases.	🌳
S6	The side length of a square increases by 2 cm and its area increases by 20 cm². Side length of the original square = ?	⛰

S7	The L shape is divided into A (12×5 cm) and B (4×8 cm). Total area = ?	

fractionanddecimal（total 5 question）

#	Question	Difficulty
S8	$\frac{5}{6} - \frac{1}{4} = ?$	
S9	$2\frac{1}{3} \times 1\frac{1}{5} = ?$	
S10	Calculate 1.25 ÷ 0.4 = ? (write down the steps)	
S11	There are 120 pieces in a box of candy. $\frac{1}{5}$ is red, $\frac{1}{3}$ is green, and the rest are yellow. How many yellow grains are there?	
S12	|||SEP||| of a number is equal to $\frac{2}{3}$. This number = ? $\frac{1}{4}$. This number = ?	

Algebraic equation＋circle＋3D shape (total 4 questions)

#	Question	Difficulty
S13	Simplify: $5(3a - 2) - 3(2a + 1) = ?$	
S14	The radius of a circle is 14 cm ($\pi \approx 22/7$). Find (a) the circumference of the circle (b) the area of the circle.	
S15	Rectangular water tank 60×40×30 cm. Fill it with water and pour it into a cube water tank (side length 40 cm). Will the water overflow? How high is the water level?	

S16	The cube water tank has a side length of 20 cm and contains $\frac{3}{4}$ water. A stone with a volume of 1500 cm^3 is thrown into it. Will the water overflow?	⛰

📝 **Post-exam review record:**

Part A:____/30 Part B:____/40 Part C:____/30 Total score:____/100

Simulation 1→2→4 Progress Curve: ____→____→____ Most in need of strengthening: __________